Chapters 3–4, condensed (English / Chinese)
第 3–4 章精简中英对照版

Condensed from the dissertation. Definitions, key equations and design decisions only; derivations, justifications and experimental evidence are omitted (see the full text). Equation numbers are local to this document.

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3 Modeling and Parameter Structure　建模与参数结构

A grasped parcel changes both the mass and the roll/pitch inertia. For a payload on the thrust axis both are governed by one online scalar: the composite mass m_T, identified from the translational channel. The inertia increment follows from m_T and a nominal gripper arm. No payload mass is given to either optimiser, and the estimator receives no attach or release notification. The estimator and controller models built on this chapter are collected in Section 4.2.

抓取包裹会同时改变质量和滚转/俯仰惯量。对于位于推力轴上的载荷，二者由同一个在线标量决定：复合质量 m_T，由平动通道辨识。惯量增量由 m_T 和夹爪的名义力臂算出。两个优化器都不提供载荷质量，估计器也不接收吸附或释放通知。基于本章的估计器与控制器模型集中放在第 4.2 节。

3.1 System definition and assumptions　系统定义与假设

World frame ENU (gravity along $-z$); body frame FLU with origin at the rotor-disc centre, not at the centre of mass.

世界系为 ENU（重力沿 $-z$）；机体系为 FLU，原点在旋翼盘几何中心，而不在质心。

$$x = [\boldsymbol{p}; \boldsymbol{v}; \boldsymbol{q}; \boldsymbol{\omega}] \in \mathbb{R}^{13}, \qquad u = [T; \boldsymbol{\tau}] \in \mathbb{R}^4, \qquad m_T = m_B + m_P. \tag{3.1}$$

Assumptions: (1) the bare airframe is balanced about the body origin; (2) the payload is a point mass rigidly attached on the thrust axis, $\boldsymbol{r}_p = [0, 0, r_z]^\top$, $r_z < 0$; (3) rotor thrust follows the static quadratic law. Attitude: unit quaternion, Hamilton convention, scalar first, $\boldsymbol{\omega}$ in the body frame. The vertical thrust fraction is

假设：(1) 空机关于机体原点平衡；(2) 载荷是刚性挂在推力轴上的质点，$\boldsymbol{r}_p = [0, 0, r_z]^\top$，$r_z < 0$；(3) 旋翼推力服从静态平方律。姿态用单位四元数，Hamilton 约定、标量在前，$\boldsymbol{\omega}$ 在机体系表示。推力的竖直分量比例为

$$\cos\phi\cos\theta = \boldsymbol{e}_3^\top R(\boldsymbol{q})\boldsymbol{e}_3 = 1 - 2(q_x^2 + q_y^2). \tag{3.2}$$

3.2 Dynamics　动力学

$$\dot{\boldsymbol{p}} = \boldsymbol{v}, \qquad \dot{\boldsymbol{q}} = \tfrac{1}{2}\,\boldsymbol{q} \otimes [0, \boldsymbol{\omega}] = \tfrac{1}{2}\,\Xi(\boldsymbol{q})\boldsymbol{\omega}, \tag{3.3}$$

$$\dot{\boldsymbol{v}} = \frac{1}{m_T}\big(R(\boldsymbol{q})\boldsymbol{e}_3 T - k_d \boldsymbol{v}\big) - g\boldsymbol{e}_3, \tag{3.4}$$

$$\dot{\boldsymbol{\omega}} = J_T^{-1}\big(\boldsymbol{\tau} - \boldsymbol{\omega} \times J_T\boldsymbol{\omega}\big). \tag{3.5}$$

The kinematics contain no physical parameter, and $\boldsymbol{q}^\top\Xi(\boldsymbol{q}) = 0$ preserves $\|\boldsymbol{q}\|$. The mass enters only as the gain $1/m_T$ in (3.4); the inertia increment only in-

运动学不含物理参数，且 $\boldsymbol{q}^\top\Xi(\boldsymbol{q}) = 0$ 保持 $\|\boldsymbol{q}\|$ 不变。质量只以增益 $1/m_T$ 出现在 (3.4)；惯量增量只出现在 (3.5) 的 J_T 中，并取关于复合质心。投影到 $\boldsymbol{e}_3$、忽略

side J_T in (3.5), taken about the composite centre of mass. Projected on $\boldsymbol{e}_3$, with vertical drag neglected:

竖直阻力：

$$m_T(g + a_z) = T\cos\phi\cos\theta. \tag{3.6}$$

3.3 Mass–inertia coupling 质量–惯量耦合

The composite centre of mass moves to $\boldsymbol{c} = (m_P/m_T)\boldsymbol{r}_p$, so the parallel-axis theorem is applied about that point and both bodies contribute. The two terms collapse onto the reduced mass:

复合质心移到 $\boldsymbol{c} = (m_P/m_T)\boldsymbol{r}_p$，因此平行轴定理要对这一点应用，两个刚体都有贡献。两项合并后正好是折合质量：

$$\mu = \frac{m_B m_P}{m_T}, \qquad \Delta J = m_B\Big(\frac{m_P}{m_T}d\Big)^2 + m_P\Big(\frac{m_B}{m_T}d\Big)^2 = \mu r_z^2, \qquad J_T = J_B + \mathrm{diag}(\mu r_z^2,\ \mu r_z^2,\ 0). \tag{3.7}$$

The increment is isotropic in roll/pitch, diagonal, and zero in yaw. The naive $m_P r_z^2$ overestimates it by m_T/m_B (14.5 % at 0.30 kg). With $\kappa = (J_{xx} + \Delta J)/J_{xx}$, $m_B = 2.0643\,\mathrm{kg}$, $J_{xx} = 0.0142\,\mathrm{kg\,m^2}$, $r_z = -0.27\,\mathrm{m}$:

增量在滚转/俯仰上各向同性、为对角阵、偏航分量为零。朴素写法 $m_P r_z^2$ 会高估 m_T/m_B 倍（0.30 kg 时高估 14.5 %）。记 $\kappa = (J_{xx} + \Delta J)/J_{xx}$，取 $m_B = 2.0643\,\mathrm{kg}$、$J_{xx} = 0.0142\,\mathrm{kg\,m^2}$、$r_z = -0.27\,\mathrm{m}$：

	μ [kg]	ΔJ [kg m^2]	κ
$m_P = 0.15\,\mathrm{kg}$	0.140	0.0102	1.72
$m_P = 0.30\,\mathrm{kg}$	0.262	0.0191	2.34

A 15 % mass change is a more than two-fold inertia change; mass adaptation alone does not cover it.

质量变化 15 %，惯量却变成两倍多；只做质量自适应覆盖不了后者。

3.4 Single-parameter estimation structure 单参数估计结构

The mass changes within a flight and is observable through $1/m_T$; the arm is a property of the gripper and is held nominal. The inertia is therefore derived, never a decision variable, and vanishes by itself at release:

质量在飞行中会变，且可通过 $1/m_T$ 观测；力臂是夹爪的属性，取名义值。因此惯量是推出来的，从不作为决策变量，释放后自动归零：

$$m_P = (m_T - m_B)^+, \qquad \boxed{\Delta J(m_T) = \frac{m_B(m_T - m_B)^+}{m_T}\, r_z^2} \tag{3.8}$$

The rotational model of this platform has three known errors (airframe inertia, rotor-drag moment, torque bias). The **deployed estimator therefore does not integrate the rotational dynamics at all**: it keeps only the translational dynamics (3.4), takes the measured attitude as a known input, and has velocity and mass as its only states (Section 4.2). No inertia, torque or arm enters it.

该平台的转动模型有三处已知误差（空机惯量、旋翼阻力力矩、力矩偏置）。因此**部署的估计器完全不积分转动动力学**：只保留平动动力学 (3.4)，把测得的姿态当作已知输入，状态只有速度和质量（见第 4.2 节）。惯量、力矩、力臂都不进入估计器。

The evaluation campaigns were flown with an earlier *full-state* estimator that does integrate (3.3) and (3.5). It keeps the mass off the rotational model only directly, inserting the increment as a known parameter from the previous solution $\hat{m}^-$ ("lagged"):

评估实验用的是更早的*全状态*估计器，它会积分 (3.3) 和 (3.5)。它只是不让质量*直接*作用于转动模型：惯量增量用上一次解 $\hat{m}^-$ 算出后作为已知参数代入（"滞后"处理）：

$$\left.\frac{\partial \dot{\boldsymbol{\omega}}}{\partial m_T}\right|_{\text{MHE}} = 0, \qquad \Delta J^{\text{lag}} = \frac{m_B(\hat{m}^- - m_B)^+}{\hat{m}^-}\, r_z^2 \;\; \text{(constant within the window / 窗内为常数).} \tag{3.9}$$

Treatment 处理	ΔJ in MHE	$\partial\dot{\boldsymbol{\omega}}/\partial m_T$	Status 状态
omitted 省略	0	0	available, off 可用，关闭
lagged 滞后	(3.8) at $\hat{m}^-$	0	campaigns 评估实验
live 实时	(3.8) at m_T	$\neq 0$	rejected 弃用
not integrated 不积分 (reduced)	none 无	—	deployed 部署默认

In the full-state estimator rotational error still reaches the mass through the estimated attitude (which sets the thrust direction), and the lagged mass feeds the next window. The reduced estimator removes both paths: it has no estimated attitude and no inertia.

在全状态估计器里，转动误差仍可经估计姿态（它决定推力方向）传到质量，滞后的质量也会影响下一个窗口。降阶估计器把这两条路径都去掉了：它没有估计姿态，也没有惯量。

Parameter 参数	Identity 身份	Value 取值
m_T	**MHE decision variable** 估计器决策变量	—
ΔJ, m_P	derived from m_T; not in the deployed estimator 由 m_T 推出；不进入部署的估计器	(3.8)
r_z	nominal gripper geometry 名义夹爪几何	$-0.27\,\text{m}$
$\boldsymbol{c}_{xy}$	not modelled, held at 0 不建模，取 0	—
m_B, J_B	calibration 标定	2.0643 kg; diag(0.0142, 0.0142, 0.0210)
k_f, k_m	calibration 标定	$8.55 \times 10^{-6}\,\text{N s}^2$, 0.016 m
k_d, g	constants 常数	0.05, 9.81

Information boundary. No task-specific payload mass enters the estimator or the controller, and no attach/release notification enters the estimator. Two platform constants remain: the nominal arm r_z and the 0.30 kg payload envelope used for the attach-time inertia floor.

信息边界。估计器和控制器都不接收任务相关的载荷质量，估计器不接收吸附/释放通知。保留的两个平台常数：名义力臂 r_z，以及吸附时惯量地板所用的 0.30 kg 载荷包线。

3.5 Measured input and interfaces 测量输入与接口

The wrench is reconstructed from measured rotor speeds (X frame, $l = 0.174\,\text{m}$):

力/力矩由实测转速重建（X 型布局，$l = 0.174\,\text{m}$）：

$$\begin{bmatrix} T_{\text{phys}} \\ \boldsymbol{\tau}_{\text{phys}} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ -l & l & l & -l \\ -l & l & -l & l \\ -k_m & -k_m & k_m & k_m \end{bmatrix}}_{M} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}, \qquad F_i = k_f \omega_i^2. \tag{3.10}$$

Rotor speeds are sampled at 250 Hz and averaged per 10 Hz interval. The deployed estimator uses only the first row, T_{phys}; the full-state estimator uses all four. The commanded input cannot be used: with $T_{\text{cmd}} = \tilde{m}g$, the estimate returns $\tilde{m}$ for any $\tilde{m}$ and carries no information.

转速以 250 Hz 采样，在每个 10 Hz 估计间隔内取平均。部署的估计器只用第一行 T_{phys}，全状态估计器四行都用。不能用指令输入：若 $T_{\text{cmd}} = \tilde{m}g$，估计结果对任意 $\tilde{m}$ 都等于 $\tilde{m}$，不含任何信息。

Autopilot interface. Three torques: the *predicted* torque (decision variable, never leaves the solver); the *body-rate setpoint* sent to PX4 at 50 Hz, interpolated along the predicted trajectory and scaled for the payload inertia; the *physical* torque from (3.10), the only one the full-state estimator sees (the deployed estima-

与飞控的接口。三种力矩：*预测力矩*（决策变量，不出求解器）；*角速度设定值*，以 50 Hz 发给 PX4，沿预测轨迹插值并按载荷惯量缩放；*物理力矩*，由 (3.10) 得到，是全状态估计器唯一看到的力矩（部署的估计器不看任何力矩）。

tor sees no torque).

3.6 Assumptions and limits 假设与局限

- **Exact:** no thrust moment for an on-axis load; μr_z^2 exact for point masses; zero yaw increment.
- **Centred grasp:** a residual offset r_{xy} gives trim $m_P g r_{xy}$ (0.09 N m at 3 cm, 0.30 kg), absorbed by the PX4 rate integrator.
- **Nominal arm:** realised 0.25 m to 0.30 m; ΔJ error -14 to $+23$ %.
- **Parcel inertia** neglected ($\approx$2 % of ΔJ).
- **Airframe rotational model** approximate: J_{xx} is 60 % of the simulator value; rotor-drag moment missing; torque bias $\sim$0.01 N m — the reason the deployed estimator does not integrate it.
- **Lag** (full-state estimator only) of one 0.1 s solve; $\dot{J}_T\boldsymbol{\omega}$ omitted.
- **Static rotor model:** an error in k_f passes one-for-one into $\hat{m}$; the deployed estimator flags it only when the estimate leaves the health band, and cannot correct it.

- **精确成立：**轴上载荷不产生推力力矩；质点情形 μr_z^2 精确；偏航增量为零。
- **居中抓取：**残余偏心 r_{xy} 产生配平力矩 $m_P g r_{xy}$（3 cm、0.30 kg 时 0.09 N m），由 PX4 速率环积分项吸收。
- **名义力臂：**实际为 0.25 m to 0.30 m, ΔJ 误差 -14 至 $+23$ %。
- **包裹自身惯量**忽略（约为 ΔJ 的 2 %）。
- **空机转动模型**近似：J_{xx} 只有仿真模型值的 60 %；缺旋翼阻力力矩；力矩偏置约 0.01 N m——这正是部署的估计器不积分转动动力学的原因。
- **滞后**（仅全状态估计器）一个 0.1 s 求解周期；忽略 $\dot{J}_T\boldsymbol{\omega}$。
- **静态旋翼模型：**k_f 的相对误差一比一传到 $\hat{m}$；部署的估计器只在估计值越出健康带时报警，无法纠正。

4 The MHE–NMPC Framework MHE–NMPC 框架

4.1 Architecture 系统架构

The only inputs are odometry and rotor speeds. The estimator publishes one atomic frame (mass, derived inertia, evidence score, health), read by both the NMPC model and the inner-loop scaling. Rates: MHE 10 Hz over a 2 s window; NMPC 20 Hz over a 1 s preview; setpoints at 50 Hz.

输入只有里程计和转速。估计器发布一个原子帧（质量、推出的惯量、证据分数、健康位），NMPC 模型和内环缩放都只读这一帧。频率：MHE 10 Hz、窗口 2 s；NMPC 20 Hz、预测 1 s；设定值 50 Hz。

4.2 Estimator and controller models 估计器与控制器模型

The three models of one plant are collected here. R is the deployed estimator, F the full-state estimator used in the evaluation campaigns, and the NMPC model is the controller's prediction model. R and F are selected by one switch (entry points default to `reduced` since 5 Nov 2026, library default `full`) and share every interface.

这里把同一被控对象的三个模型放在一起：R 是部署的估计器，F 是评估实验所用的全状态估计器，NMPC 模型是控制器的预测模型。R 和 F 由一个开关选择（自 2026-11-05 起入口脚本默认 `reduced`,库默认仍为 `full`），两者共用全部接口。

Switch / 开关：`MHE_MODEL_VARIANT=reduced|full`

	R (deployed 部署)	**F** (campaigns 评估实验)	**NMPC**
State 状态	$[\boldsymbol{v}; m_T] \in \mathbb{R}^4$	$[\boldsymbol{p}; \boldsymbol{v}; \boldsymbol{q}; \boldsymbol{\omega}; m_T] \in \mathbb{R}^{14}$	$[\boldsymbol{p}; \boldsymbol{v}; \boldsymbol{q}; \boldsymbol{\omega}] \in \mathbb{R}^{13}$
Known inputs 已知输入	$\tilde{\boldsymbol{q}}$, T_{phys} ($\mathbb{R}^5$)	$T_{\text{phys}}, \boldsymbol{\tau}_{\text{phys}}, \Delta J^{\text{lag}}, \boldsymbol{c}_{xy}$=0 ($\mathbb{R}^7$)	$x_{\text{ref}}, m_T, \Delta J, \boldsymbol{c}_{xy}$=0, $\hat{\boldsymbol{d}}$ ($\mathbb{R}^{20}$)
Rotational dyn. 转动动力学	not integrated 不积分	integrated, $\partial\dot{\boldsymbol{\omega}}/\partial m_T = 0$	integrated with $J_T(\Delta J)$
Residual 残差	velocity 速度 $\boldsymbol{e}^v \in \mathbb{R}^3$	full state 全状态 $\boldsymbol{e} \in \mathbb{R}^{13}$	tracking + input 跟踪 + 输入
Rate 频率	10 Hz, $N = 20$, 0.1 s	same 同	20 Hz, $N = 20$, 0.05 s

R (deployed). Velocity and mass are the only states; the measured attitude and the interval-averaged thrust are known parameters:

R（部署）。状态只有速度和质量；测得的姿态和区间平均推力作为已知参数：

$$
\begin{aligned}
x_{\text{R}} = [\boldsymbol{v}; m_T] \in \mathbb{R}^4, \qquad p_{\text{R}} = [\tilde{\boldsymbol{q}}; T_{\text{phys}}], \quad \tilde{\boldsymbol{q}} = \frac{\boldsymbol{q}^{\text{meas}}}{\|\boldsymbol{q}^{\text{meas}}\|}; \\
\dot{\boldsymbol{v}} = \frac{1}{m_T}\Big(\frac{R(\tilde{\boldsymbol{q}})\boldsymbol{e}_3}{\|\tilde{\boldsymbol{q}}\|^2} T_{\text{phys}} - k_d \boldsymbol{v}\Big) - g\boldsymbol{e}_3 + \boldsymbol{w}, \quad \dot{m}_T = 0; \qquad \boldsymbol{e}_j^v = \boldsymbol{y}_j^v - \boldsymbol{v}_j.
\end{aligned}
\tag{4.1}
$$

$R(\boldsymbol{q})$ is quadratic in $\boldsymbol{q}$, so dividing by $\|\tilde{\boldsymbol{q}}\|^2$ normalises inside the model, and $R(\boldsymbol{q}) = R(-\boldsymbol{q})$ needs no hemisphere alignment. The attitude is an input: no quaternion residual, no norm penalty. Since the mass enters the plant only as $1/m_T$ in (3.4), the velocity response to a known thrust direction carries all the window's information about it; supplying the measured attitude removes the path by which rotational model error reached the mass in F. Price: attitude noise enters the thrust direction unfiltered (untested, as SITL odometry is near-ideal).

$R(\boldsymbol{q})$ 是 $\boldsymbol{q}$ 的二次型，除以 $\|\tilde{\boldsymbol{q}}\|^2$ 就在模型内部完成归一化，且 $R(\boldsymbol{q}) = R(-\boldsymbol{q})$，无需半球对齐。姿态是输入而不是状态：没有四元数残差，也没有模长代价。质量只以 $1/m_T$ 出现在 (3.4) 中，所以给定推力方向和大小后，速度响应已包含窗口内关于质量的全部信息；直接用测得的姿态，就切断了 F 中转动模型误差传到质量的路径。代价：姿态测量噪声不经滤波进入推力方向（仿真里程计近乎理想，未经实验检验）。

F (campaigns). The mass is augmented to the 13-state vehicle model; process noise on the 13 physical states only; the residual compares velocities, never accelerations:

F（评估实验）。质量增广到 13 维飞行器模型上；过程噪声只加在 13 个物理状态上；残差比较的是速度而不是加速度：

$$x_{\text{aug}} = [x; m_T] \in \mathbb{R}^{14},\ \dot{m_T} = 0; \qquad p_{\text{MHE}} = [T_{\text{phys}}; \boldsymbol{\tau}_{\text{phys}}; \Delta J^{\text{lag}}; c_x; c_y] \in \mathbb{R}^7; \qquad \boldsymbol{e}_i = \boldsymbol{y}_i - x_i = [\boldsymbol{e}^p; \boldsymbol{e}^v; \boldsymbol{e}^q; \boldsymbol{e}^\omega]_i. \tag{4.2}$$

NMPC. m_T is the published (filtered) estimate, ΔJ is (3.8) at it, $\boldsymbol{c}_{xy} = 0$, and $\hat{\boldsymbol{d}}$ is the residual specific force. The controller is the only model that needs the inertia, and it derives it from the estimate:

NMPC。m_T 取发布的（滤波后）估计，ΔJ 为该值代入 (3.8)，$\boldsymbol{c}_{xy} = 0$，$\hat{\boldsymbol{d}}$ 为残余比力。三个模型中只有控制器需要惯量，而且它从估计值推出惯量：

$$p_{\text{NMPC}} = [x_{\text{ref}}; m_T; \Delta J; \boldsymbol{c}_{xy}; \hat{\boldsymbol{d}}] \in \mathbb{R}^{20}, \qquad \dot{\boldsymbol{v}} = \frac{1}{m_T}\big(R\boldsymbol{e}_3 T - k_d \boldsymbol{v}\big) - g\boldsymbol{e}_3 + \hat{\boldsymbol{d}}, \qquad J_T = \operatorname{diag}(J_{xx} + \Delta J, J_{yy} + \Delta J, J_{zz}). \tag{4.3}$$

4.3 Moving horizon estimator 滑动时域估计器

Both estimators solve the same problem; F_M is one RK4 step of the respective model and C selects the velocity (R) or the 13 vehicle states (F):

两种估计器求解同一形式的问题；F_M 是各自模型的一步 RK4，C 选出速度（R）或 13 个飞行器状态（F）：

$$\begin{aligned} \min_{x_{0:N}, w_{0:N-1}} \ & \|x_0 - \bar{x}_0\|^2_{W_0} + \sum_{j=0}^{N-1} \beta_j \big(\|y_j - C x_j\|^2_{W_y} + \|w_j\|^2_{W_w} \big) \\ \text{s.t. } & x_{j+1} = F_M(x_j, p_j) + G w_j, \quad G = \begin{bmatrix} I_{n_w} \\ 0 \end{bmatrix} \ (n_w = 3 \text{ R},\ 13 \text{ F}), \quad \underline{m} \le m_T \le \overline{m}. \end{aligned} \tag{4.4}$$

$N = 20$, $\Delta t = 0.1\,\text{s}$; stage j pairs the measurement at the start of the interval with the input averaged over it. Weights by Bryson's rule (F):

$N = 20$，$\Delta t = 0.1\,\text{s}$；第 j 段把区间起点的测量与该区间平均输入配对。权重按 Bryson 规则（F）：

$$\begin{gathered} W_y = \operatorname{diag}(\sigma_p^{-2} I_3, \sigma_v^{-2} I_3, \sigma_q^{-2} I_4, \sigma_\omega^{-2} I_3), \quad \sigma_p = 0.02\,\text{m},\ \sigma_v = 0.05\,\text{m s}^{-1},\ \sigma_q = 0.01,\ \sigma_\omega = 0.02\,\text{rad s}^{-1}; \\ W_w = \operatorname{diag}(10^4 I_6, 10^5 I_4, 10^3 I_3); \qquad W_0 = \operatorname{diag}(10^3 I_6, 10^4 I_4, 10^2 I_3, \sigma_{m,0}^{-2}),\ \sigma_{m,0} = 3.16\,\text{kg}; \\ \text{R: } W_y = \sigma_v^{-2} I_3, \quad W_w = 10^4 I_3, \quad W_0 = \operatorname{diag}(10^3 I_3, \sigma_{m,0}^{-2}). \end{gathered} \tag{4.5}$$

R uses exactly the velocity and mass blocks of F's weights, so the evidence for the mass is weighted identically. Process noise is tighter than measurement noise so that persistent thrust–motion disagreement is explained by the mass; the mass anchor is almost uninformative but non-zero (zero caused 472 solver failures).

R 的权重正是 F 权重中速度和质量那几块，因此两者对质量证据的加权完全相同。过程噪声权重比测量权重更紧，迫使推力与运动之间的持续不符由质量来解释；质量锚点几乎不含信息但不为零（设为零曾导致 472 次求解失败）。

Mass bounds and health check. F: hard bound $[0.95 m_B, 5\,\text{kg}]$; a $-10\,\%$ thrust map needs $0.9 m_B = 1.858\,\text{kg}$ and is pinned at the bound (replay: 107 solves ended at the iteration limit, all at $-10\,\%$). R separates the numerical bound, $[0.70 m_B, 5\,\text{kg}]$, from a physical *health band* checked after each solve:

质量边界与健康检查。F：硬边界 $[0.95 m_B, 5\,\text{kg}]$；$-10\,\%$ 推力映射要求 $0.9 m_B = 1.858\,\text{kg}$，会被钉在边界上（回放中 107 次求解跑满迭代，全部在 $-10\,\%$ 工况）。R 把数值边界 $[0.70 m_B, 5\,\text{kg}]$ 与物理*健康带*分开，每次求解后检查：

$$m_B - 0.10\,\text{kg} \le \hat{m} \le m_B + m_P^{\text{env}} + 0.10\,\text{kg} = [1.964\,\text{kg},\ 2.464\,\text{kg}]. \tag{4.6}$$

Below: possible thrust-map under-estimate; above: out of envelope or over-estimate; also flagged near the numerical box or non-finite. The flag acts on the first offending solve and clears after 10 in-band solves. A flagged estimate is not forwarded: the frame is unhealthy and the controller keeps its last accepted model. Margins from the 312-flight replay (nominal estimates

低于下限：推力映射可能偏小；高于上限：超出包线或推力映射偏大；贴近数值边界或非有限值也会报。第一次越界即生效，连续 10 次回到带内才恢复。被标记的估计不转发：帧标为不健康，控制器保持上一帧接受的模型。阈值来自 312 架回放（名义推力下估计范围 1.985 kg to 2.368 kg），不使用仿真真值。健康检查只能发现、不能纠正：仅凭平动方程只能看到 T_{phys}/m_T，所以不同时估计

1.985 kg to 2.368 kg); no simulator truth used. The check detects but cannot correct: translation alone sees only T_{phys}/m_T, so mass and thrust scale are not estimated jointly ($-10\,\%$ with 0.30 kg loaded, $\hat{m} \approx 2.13$ kg, is not detected).

质量和推力比例（$-10\,\%$ 且带 0.30 kg 时 $\hat{m} \approx 2.13$ kg，检测不到）。

Input gating and the grasp gap. The thrust is admitted only while the odometry height exceeds 0.5 m (ground support); inputs older than 0.30 s clear the window. The grasp hover at 0.35 m is below the gate, so in every inspected flight the window is cleared 1–3 s before the attach and rebuilt about 6 s later, after lift-off, from the window-balance seed. The grasp is never inside a window.

输入门控与抓取空窗。只有在里程计高度高于 0.5 m 时才接收推力输入（防地面支撑力）；输入超过 0.30 s 未更新就清空窗口。抓取悬停高度 0.35 m 低于门槛，所以在检查过的每一架里，窗口都在吸附前 1–3 s 被清空，离地后约 6 s 再用窗口力平衡种子重建。抓取过程从不落在估计窗口内。

Seed. A prior-free window force balance, using only the two velocity endpoints (no differentiation):

初值种子。不含先验的窗口力平衡，只用两个速度端点（不做数值微分）：

$$m^{(0)} = \frac{\sum_{i=0}^{N-1} T_i \cos\phi_i \cos\theta_i}{Ng + (v_{z,N} - v_{z,0})/\Delta t}, \qquad \text{fallback / 退化时} m^{(0)} = \text{clip}(T_{\text{phys}}/g). \tag{4.7}$$

It sets the initial iterate and the arrival-cost mean (weak): where the solver starts, not where it converges.

种子用作初始迭代点和到达代价均值（权重很弱）：只决定求解器从哪里出发，不决定收敛到哪里。

Stage weights. Residual-triggered de-weighting (pre-event stages $\times 10^{-4}$ after a 4.4 N thrust step) is enabled but rarely acts: its detections mostly fall inside the grasp gap, so in the 312-flight campaign it fired in 81 flights and changed the weights in 10. Weights are uniform in effect; forgetting and a mass random walk exist but are off. For R the de-weighting matrices are rebuilt on R's own blocks.

阶段权重。残差触发的降权（推力阶跃超过 4.4 N 后，事件前各段权重乘 10^{-4}）是开着的，但几乎不起作用：它的检测大多落在抓取空窗内，所以 312 架实验中触发 81 架、真正改动权重的只有 10 架。实际效果是均匀权重；遗忘因子和质量随机游走已实现但关闭。R 的降权矩阵按 R 自己的维度重建。

Solver failure. A non-zero status or non-finite iterate is discarded and the frame marked unhealthy; after five consecutive failures the window is rebuilt and re-seeded from (4.7). F adopts the seed as its estimate; R uses it only as the arrival mean, keeps the last valid estimate, and logs status, SQP iterations and failure counts.

求解失败。非零状态或非有限解一律丢弃，帧标为不健康；连续五次失败后重建窗口并用 (4.7) 重新给种子。F 直接把种子当作当前估计；R 只把种子用作到达代价均值，保持上一帧有效估计，并记录状态码、SQP 迭代数和失败计数。

4.4 Adaptive NMPC　自适应 NMPC

$$\begin{aligned} \min_{u_{0:N-1}} \; & \sum_{k=0}^{N-1} \Big(\|e_{\text{track},k}\|_Q^2 + \|u_k - u_{\text{hover}}\|_R^2 \Big) + \|e_{\text{track},N}\|_P^2 \\ \text{s.t. } & x_{k+1} = F(x_k, u_k; p_{\text{NMPC}}), \quad \underline{T} \le T_k \le \overline{T}, \quad |\tau_{k,i}| \le \overline{\tau}_i, \end{aligned} \qquad T_h = m_T(g - \hat{d}_z),\; u_{\text{hover}} = [T_h, 0, 0, 0]^\top. \tag{4.8}$$

The hover baseline is recomputed from the live estimate; a stale (unloaded) baseline pulls thrust below equilibrium and leaves a steady-state error that a larger R makes worse. $N = 20$, $\Delta t = 0.05$ s; attitude error is the vector part of $\boldsymbol{q}_{\text{ref}}^* \otimes \boldsymbol{q}$. Bryson tolerances: $L_p = 0.1$ m, $L_v = 0.3\,\text{m s}^{-1}$, $L_q = 0.05$, $L_\omega = 0.3\,\text{rad s}^{-1}$, $L_T = 2$ N, $L_{\tau,xy} = 0.5$ N m, $L_{\tau,z} = 0.2$ N m; terminal scaling 20/1/2/0.2.

悬停基准按实时估计重算；过时的（空机）基准会把推力拉到平衡值以下，留下稳态误差，而且 R 越大越严重。$N = 20$，$\Delta t = 0.05$ s；姿态误差取 $\boldsymbol{q}_{\text{ref}}^* \otimes \boldsymbol{q}$ 的向量部分。Bryson 容差：$L_p = 0.1$ m、$L_v = 0.3\,\text{m s}^{-1}$、$L_q = 0.05$、$L_\omega = 0.3\,\text{rad s}^{-1}$、$L_T = 2$ N、$L_{\tau,xy} = 0.5$ N m、$L_{\tau,z} = 0.2$ N m；终端缩放 20/1/2/0.2。

Residual specific force ($\mathcal{L}_1$-type predictor and adaptation law), compensating the translational effect of mass bias:

残余比力（$\mathcal{L}_1$ 型状态预测器与自适应律），补偿质量偏差在平动上的影响：

$$\dot{\hat{\boldsymbol{v}}} = \boldsymbol{a}_{\text{nom}} + \hat{\boldsymbol{d}}_{\text{raw}} - a(\hat{\boldsymbol{v}} - \boldsymbol{v}), \quad \dot{\hat{\boldsymbol{d}}}_{\text{raw}} = -\Gamma(\hat{\boldsymbol{v}} - \boldsymbol{v}), \quad \hat{\boldsymbol{d}} = \frac{\omega_c}{s + \omega_c}\hat{\boldsymbol{d}}_{\text{raw}}, \quad a = 10,\ \Gamma = (a/2)^2,\ \omega_c = 0.5. \tag{4.9}$$

$\boldsymbol{a}_{\text{nom}}$ uses the measured thrust and the published mass. $\hat{\boldsymbol{d}}$ is controller-only (never reaches the estimator or lifecycle), reset on an accepted commanded release, and held at zero below $0.4 m_B g$ (ground support).

$\boldsymbol{a}_{\text{nom}}$ 用实测推力和发布的质量计算。$\hat{\boldsymbol{d}}$ 只在控制器中使用（不进估计器、不参与生命周期判决），有指令的释放被接受时复位，推力低于 $0.4 m_B g$ 时（地面支撑）保持为零。

Actuator envelope. Throttle maps affinely to rotor speed ($\omega_{\min} = 150$, $\Delta\omega = 850\,\text{rad}\,\text{s}^{-1}$), clipped to $[0.05, 0.95]$:

执行器包线。油门仿射映射到转速（$\omega_{\min} = 150$，$\Delta\omega = 850\,\text{rad}\,\text{s}^{-1}$），并截断在 $[0.05, 0.95]$：

$$\eta(T) = \frac{\sqrt{T/4k_f} - \omega_{\min}}{\Delta\omega}, \qquad T \in [1.27\,\text{N},\ 31.35\,\text{N}]; \qquad \bar{\tau}_{x,y} = 0.5\,\text{N}\,\text{m},\ \bar{\tau}_z = 0.2\,\text{N}\,\text{m}. \tag{4.10}$$

A weight-based box ($2 m_B g = 40.5\,\text{N}$) overstates thrust by 29 %. Yaw costs 10.9× more differential thrust than roll, so the yaw bound is marginally optimistic. **Solver:** acados, multiple shooting, RK4, Gauss–Newton, HPIPM, full SQP (RTI diverged silently).

按重量定的推力框（$2 m_B g = 40.5\,\text{N}$）会高估 29 %。偏航所需差动推力是滚转的 10.9 倍，所以偏航力矩上限略偏乐观。**求解器：** acados，多重打靶、RK4、Gauss–Newton、HPIPM，完整 SQP（RTI 曾静默发散）。

4.5 Continuous payload interface　连续载荷接口

$$\mathcal{P} = \{m_T,\ \Delta J_{xx}, \Delta J_{yy}, \Delta J_{zz},\ J_{xx}, J_{yy}, J_{zz},\ \gamma,\ h,\ a\}, \qquad \Delta J_{zz} = 0. \tag{4.11}$$

Healthy = last solve succeeded, all inputs fresher than 0.30 s and (R) the estimate inside (4.6); the consumer also rejects frames older than 0.35 s. *An unhealthy frame freezes the model and resets the confirmation timer*: it can delay a release, never cause one. Each channel is low-passed ($\tau = 0.20\,\text{s}$) and rate-limited ($0.60\,\text{kg}\,\text{s}^{-1}$; $0.080\,\text{kg}\,\text{m}^2\,\text{s}^{-1}$) so that parameter steps become ramps the warm-started solver can follow.

健康 = 上次求解成功、所有输入新于 0.30 s，且 (R) 估计值在 (4.6) 内；消费端还会拒绝超过 0.35 s 的帧。*不健康帧冻结模型并清零确认计时*：只能推迟释放，不能触发释放。每个通道先低通（$\tau = 0.20\,\text{s}$）再限速（$0.60\,\text{kg}\,\text{s}^{-1}$；$0.080\,\text{kg}\,\text{m}^2\,\text{s}^{-1}$），把参数阶跃变成热启动求解器跟得上的斜坡。

4.6 Inner-loop scheduling　内环调度

The PX4 rate loop is tuned for the bare airframe; a payload lowers its bandwidth by κ. Instead of rewriting PX4 gains, the roll/pitch rate command is scaled about the measured rate:

PX4 速率环按空机整定，载荷会让其带宽降低 κ 倍。不改 PX4 增益，而是围绕实测角速度缩放滚转/俯仰角速度指令：

$$\boldsymbol{\omega}'_{\text{cmd}} = \boldsymbol{\omega} + \varsigma(\boldsymbol{\omega}_{\text{cmd}} - \boldsymbol{\omega}), \quad \varsigma \to \kappa\ (\text{cap } 5.0); \qquad \varsigma_{\text{eff}} = \max\Big(1,\ \min_{i \in \{x,y\}} \varsigma_i^{\max}\Big),\ |\omega + \varsigma e| \le 0.95\,\bar{\omega}. \tag{4.12}$$

Dead band 0.08 (snapping to 1 and to the cap), low-pass 0.5 s, rate limits $4.0\,\text{s}^{-1}$ up / $1.5\,\text{s}^{-1}$ down. The floor at 1 means scheduling can help but never hurt.

死区 0.08（并吸附到 1 和上限），低通 0.5 s，速率限制上升 $4.0\,\text{s}^{-1}$、回落 $1.5\,\text{s}^{-1}$。下限取 1，保证调度只会帮忙、不会帮倒忙。

Attach-time inertia bootstrap. At grasp enable the inertia alone (not the mass) is floored at the 0.30 kg envelope value, $\Delta J = 0.0191\,\text{kg}\,\text{m}^2$ ($\kappa = 2.34$); the controller uses max(floor, estimate). In the evaluated configuration the floor is held until the release command, then faded over 0.5 s. It only raises inertia, never touches the mass channel. Known failure: a missed grasp leaves the floor on an empty airframe.

吸附时惯量自举。抓取使能时只把惯量（不是质量）抬到 0.30 kg 包线对应的地板值 $\Delta J = 0.0191\,\text{kg}\,\text{m}^2$（$\kappa = 2.34$）；控制器取地板值与估计值中较大者。评估配置中地板保持到投放指令，再在 0.5 s 内淡出。它只会抬高惯量，不碰质量通道。已知失效：抓取落空时，空机会一直带着这个地板。

4.7 Grasp–release lifecycle 抓取–释放生命周期

Four things are kept apart: (1) mass, inertia and inner-loop scale follow the estimate in every state; (2) the controller owns the lifecycle state (carry / pending / accepted / UNRESOLVED); (3) the residual $\hat{\boldsymbol{d}}$ is reset on an accepted commanded release; (4) task completion coincides with acceptance. *Acceptance never sets the mass or inertia.*

四件事分开处理：(1) 质量、惯量、内环缩放在任何状态下都跟随估计；(2) 生命周期状态（吊运 / 待确认 / 已接受 / UNRESOLVED）归控制器所有；(3) 有指令的释放被接受时复位残余比力 $\hat{\boldsymbol{d}}$；(4) 任务完成与接受同时发生。*接受释放从不设置质量或惯量。*

Presence (estimator). LOADED when $m_P > 0.09\,\mathrm{kg}$ for 2 s; EMPTY when $m_P < 0.02\,\mathrm{kg}$ for 5 s in quiet flight. **Evidence score** (smooth-step σ, output low-pass 0.25 s):

有无载荷（估计器侧）。$m_P > 0.09\,\mathrm{kg}$ 持续 2 s 进入 LOADED；平稳飞行中 $m_P < 0.02\,\mathrm{kg}$ 持续 5 s 回到 EMPTY。**空载证据分数**（平滑阶跃 σ，输出再低通 0.25 s）：

$$\sigma(v;a,b) = 1-(3\chi^2-2\chi^3),\ \chi = \tfrac{|v|-a}{b-a}; \qquad \gamma^\star = \sigma(\bar{m}_P;\,0.015,\,0.060); \qquad \textbf{confirm / 确认: } \gamma \geq 0.90 \text{ held / 持续} 1.0\,\mathrm{s}. \tag{4.13}$$

γ is a deterministic score, not a probability; $\gamma^\star > 0.9$ below about 0.024 kg. Two arming policies on this evidence: *always-on* (counted through carry) and *command-armed* (counted only after the command).

γ 是确定性分数，不是概率；约 0.024 kg 以下时 $\gamma^\star > 0.9$。基于这一证据有两种武装策略：*全程开启*（吊运中一直计数）和*指令武装*（只在指令之后计数）。

unresolved. No confirmation within 12 s of the command: nothing is accepted and $\hat{\boldsymbol{d}}$ is not reset; the vehicle brakes to hover along a quintic over $T_b = 3\,\mathrm{s}$ to $\boldsymbol{p}_1 = \boldsymbol{p}_0 + \frac{1}{2}T_b\boldsymbol{v}_0$ (zero acceleration at both ends). *A timeout is not a release.*

unresolved。指令后 12 s 内未确认：不接受释放、不复位 $\hat{\boldsymbol{d}}$；飞机沿五次多项式在 $T_b = 3\,\mathrm{s}$ 内制动到 $\boldsymbol{p}_1 = \boldsymbol{p}_0 + \frac{1}{2}T_b\boldsymbol{v}_0$ 并悬停（两端加速度为零）。*超时不等于释放。*

Force-balance evidence and loss watchdog. The IMU specific force f_z contains no gravity, so $f_z \approx T/m$ gives a mass that bypasses the attitude, the velocity history and the rotational model:

力平衡证据与意外脱落看门狗。IMU 测得的比力 f_z 不含重力，由 $f_z \approx T/m$ 可得一个不经过姿态、速度历史和转动模型的质量：

$$m_{\mathrm{acc}} = \frac{\bar{T}}{\max(\bar{f}_z,\,1\,\mathrm{m\,s^{-2}})}, \quad T = \textstyle\sum_i k_f\omega_i^2; \qquad \gamma^\star_{\mathrm{acc}} = \sigma\big((m_{\mathrm{acc}} - m_B)^+;0.015,0.060\big), \tag{4.14}$$

with low-pass 0.5 s on T and f_z and 0.25 s on the score; the score starts at 0 and is held while data are staler than 0.2 s. Confirmation is the same ≥ 0.90 for 1 s, but counts only if (a) a release command is pending, or (b) the watchdog is armed: after lift start, $\gamma_{\mathrm{acc}} < 0.75$ for 2.0 s; it is never disarmed by a rising score and is cleared at the next pickup. An acceptance in case (b) is an *uncommanded loss*: zero the payload geometry, end the inertia floor, reset the inner-loop scale and $\hat{\boldsymbol{d}}$, suppress later planned releases; no gripper command. Weakness: m_{acc} scales one-for-one with a k_f error.

T 和 f_z 低通 0.5 s，分数低通 0.25 s；分数初值为 0，数据比 0.2 s 更旧时保持不变。确认条件同为 ≥ 0.90 持续 1 s，但只有在以下情况才计数：(a) 有待确认的投放指令；或 (b) 看门狗已武装——起吊开始后 $\gamma_{\mathrm{acc}} < 0.75$ 持续 2.0 s；武装后不因分数上升而撤销，下次抓取时才清除。情况 (b) 下的接受判为*意外脱落*：几何归零、结束惯量地板、复位内环缩放和 $\hat{\boldsymbol{d}}$、取消之后的计划投放；不发夹爪指令。弱点：m_{acc} 随 k_f 误差一比一偏移。

Limit of a mass-only lifecycle. In manoeuvre a low mass bias can push $\hat{m}_P$ below the threshold while loaded (observed with F; smaller with R, but the argument does not depend on its size); no rule that admits only $\hat{m}$ can tell this from a release. Command-arming avoids it by not counting before the command; the force-balance rule avoids it by not using $\hat{m}$.

仅靠质量的生命周期的局限。机动中质量低偏可能让带载时的 $\hat{m}_P$ 跌破阈值（在 F 上观察到；R 的偏差更小，但论证不依赖偏差大小）；只看 $\hat{m}$ 的规则无法把它和真实释放区分开。指令武装靠指令前不计数来避开；力平衡规则靠不使用 $\hat{m}$ 来避开。

Threshold 阈值	Value 取值	Provenance 来源
presence entry / exit 进入/退出	$> 0.09\,\mathrm{kg}$, 2 s / $< 0.02\,\mathrm{kg}$, 5 s, quiet	engineering choice 工程选择
evidence score 证据分数	$\gamma \geq 0.90$, 1.0 s	engineering choice 工程选择
force-balance score 力平衡分数	LPF 0.5 s/0.25 s, fresh 0.2 s	checked by offline replay, frozen before the campaign 离线回放核对，批次前冻结
loss watchdog 看门狗	$\gamma_{\mathrm{acc}} < 0.75$, 2.0 s, after lift	frozen before the campaign 批次前冻结
UNRESOLVED / brake 制动	12 s / 3 s	engineering choice 工程选择

4.8 Reference trajectory 参考轨迹

$$x_r = r\sin a, \quad y_r = \tfrac{r}{2}\sin 2a, \quad z_r = z_h + d_z \sin a, \quad a = wt; \qquad \|\boldsymbol{v}\|_{\max} = \sqrt{2}\, rw. \tag{4.15}$$

Yaw follows the velocity. Amplitude and heading are blended in with the quintic smooth step $10s^3 - 15s^4 + 6s^5$, ramp $\geq 2.65/w$.

偏航沿速度方向。幅值和航向用五次平滑阶跃 $10s^3 - 15s^4 + 6s^5$ 逐步引入，斜坡时长 $\geq 2.65/w$。

4.9 Execution sequence 在线执行流程

1. odometry in; wrench from rotor speeds (3.10);
2. append to the window, re-seed if rebuilt; solve (4.4); (R) health check (4.6);
3. update presence and γ; publish the frame;
4. controller: check freshness, filter, write m_T, ΔJ, u_{hover};
5. evaluate confirmation / timeout / brake;
6. warm-start and solve (4.8); publish at 50 Hz with throttle inversion and rate scaling.

Estimator failure or an out-of-band estimate freezes the model; controller failure falls back to a safe action.

1. 读入里程计；由转速重建力/力矩 (3.10)；
2. 加入窗口，重建时重新给种子；求解 (4.4)；（R）健康检查 (4.6)；
3. 更新有无载荷判断和 γ；发布帧；
4. 控制器: 检查新鲜度、滤波, 写入 m_T、ΔJ、u_{hover}；
5. 判断确认 / 超时 / 制动；
6. 热启动求解 (4.8)；以 50 Hz 发布，做油门反算和角速度缩放。

估计器失败或估计越出健康带则冻结模型；控制器失败则退回安全动作。